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Course : Software Engineering

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Experiment – 14

Question :

Demonstrate how to work collaboratively in Git/GitHub on a project using the fork-and-pull request workflow.

Tasks:

- 1. Fork an existing public GitHub repository (e.g., a sample JavaScript or Python project).**
- 2. Clone the forked repository to your local machine using Git.**
- 3. Create a new branch for the feature or change you want to work on.**
- 4. Make modifications or add new features (e.g., add a function, fix a bug, or update the README).**
- 5. Commit your changes and push the branch to GitHub.**
- 6. Go to the GitHub repository and create a pull request to merge your feature branch into the main branch.**
- 7. Review the pull request and provide feedback on the changes.**
- 8. Respond to feedback by making additional commits to the feature branch if necessary.**
- 9. Once the pull request is approved, merge it into the main branch.**
- 10. Finally submit the link to the pull request along with a summary of the changes you made and how you collaborated.**

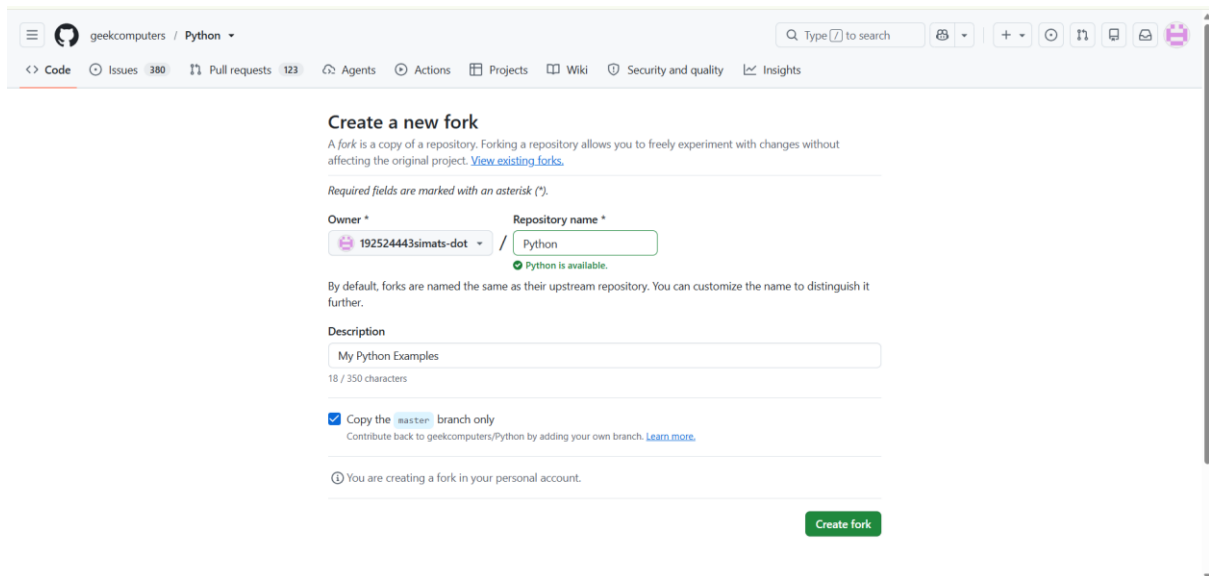
Git Hub Repository Link : <https://github.com/tamilelakiya07/Python>

Aim

To demonstrate collaborative development using Git and GitHub through the fork-and-pull request workflow.

Procedure

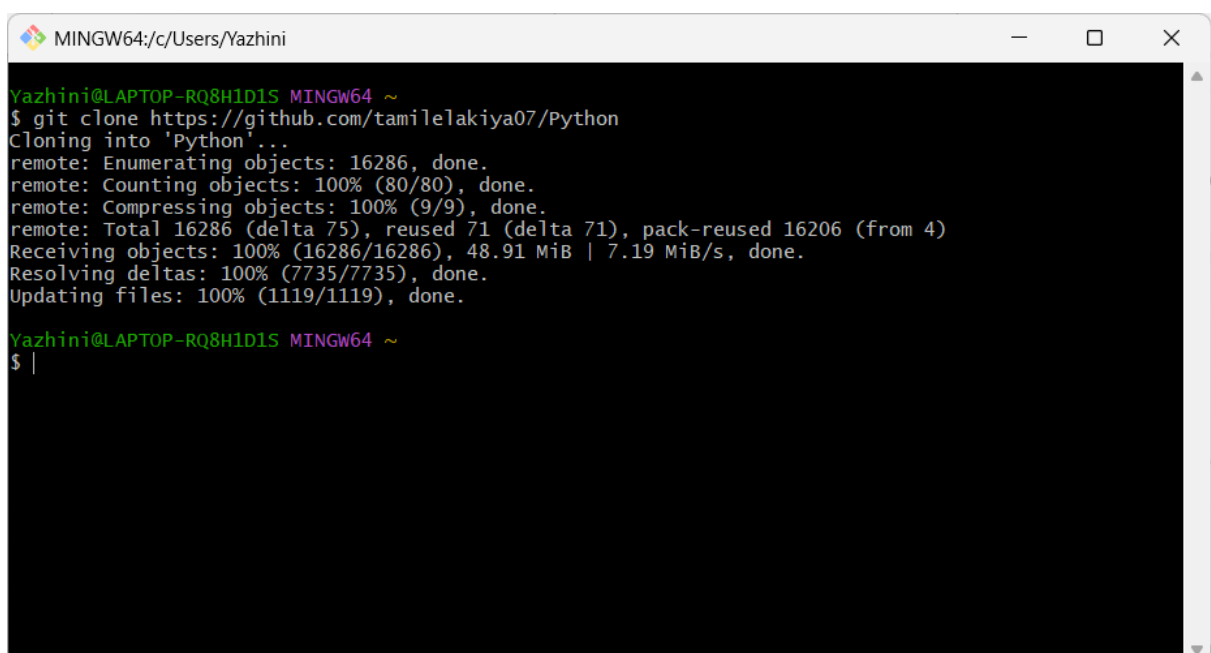
1. **Fork:** Forked an existing public GitHub repository into my GitHub account.



The screenshot shows the GitHub interface for creating a new fork of the 'Python' repository by 'geekcomputers'. The page title is 'Create a new fork'. Below the title, a brief explanation states: 'A fork is a copy of a repository. Forking a repository allows you to freely experiment with changes without affecting the original project. [View existing forks.](#)'. A note indicates 'Required fields are marked with an asterisk (*)'. The form includes an 'Owner' dropdown set to '192524443simats-dot', a 'Repository name' input field containing 'Python' with a green checkmark and the text 'Python is available.', and a 'Description' text area with 'My Python Examples' and a character count of '18 / 350 characters'. A checkbox labeled 'Copy the master branch only' is checked, with a sub-note 'Contribute back to geekcomputers/Python by adding your own branch. [Learn more.](#)'. At the bottom, a note says 'You are creating a fork in your personal account.' and a green 'Create fork' button is visible.

2. **Clone:** Cloned the forked repository to the local machine using Git Bash.

git clone <repository-url>



The screenshot shows a Windows Command Prompt window titled 'MINGW64:c/Users/Yazhini'. The user 'Yazhini@LAPTOP-RQ8H1D1S' is in the 'MINGW64' environment. They have executed the command 'git clone https://github.com/tamilelakiya07/Python'. The output shows the cloning process: 'Cloning into 'Python'...', 'remote: Enumerating objects: 16286, done.', 'remote: Counting objects: 100% (80/80), done.', 'remote: Compressing objects: 100% (9/9), done.', 'remote: Total 16286 (delta 75), reused 71 (delta 71), pack-reused 16206 (from 4)', 'Receiving objects: 100% (16286/16286), 48.91 MiB | 7.19 MiB/s, done.', 'Resolving deltas: 100% (7735/7735), done.', and 'Updating files: 100% (1119/1119), done.'. The prompt is now ready for the next command.

3. **Branch:** Created a separate feature branch.

git checkout -b update-readme

```
PS E:\Software engineering\Python> git branch
* master
PS E:\Software engineering\Python> git checkout -b update-readme
>>
Switched to a new branch 'update-readme'
PS E:\Software engineering\Python> git branch
master
* update-readme
PS E:\Software engineering\Python> git status
On branch update-readme
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md
```

4. **Modify:** Updated the README.md file with additional project documentation.

5. **Commit:** Staged and committed the changes.

git add .

git commit -m "Update README documentation"

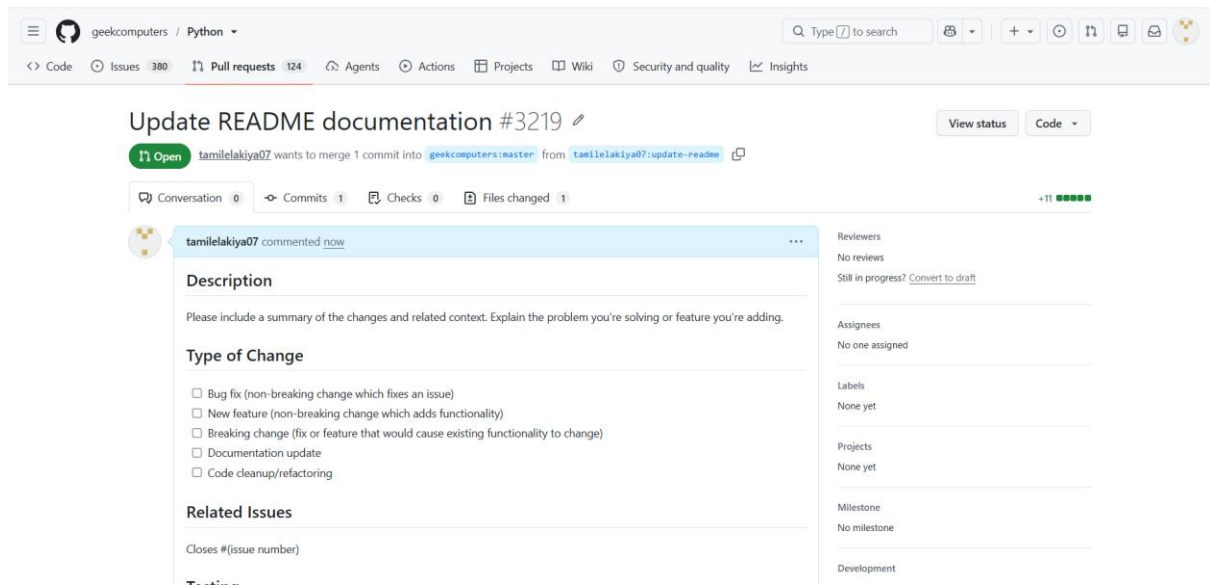
6. **Push:** Pushed the feature branch to GitHub.

git push -u origin update-readme

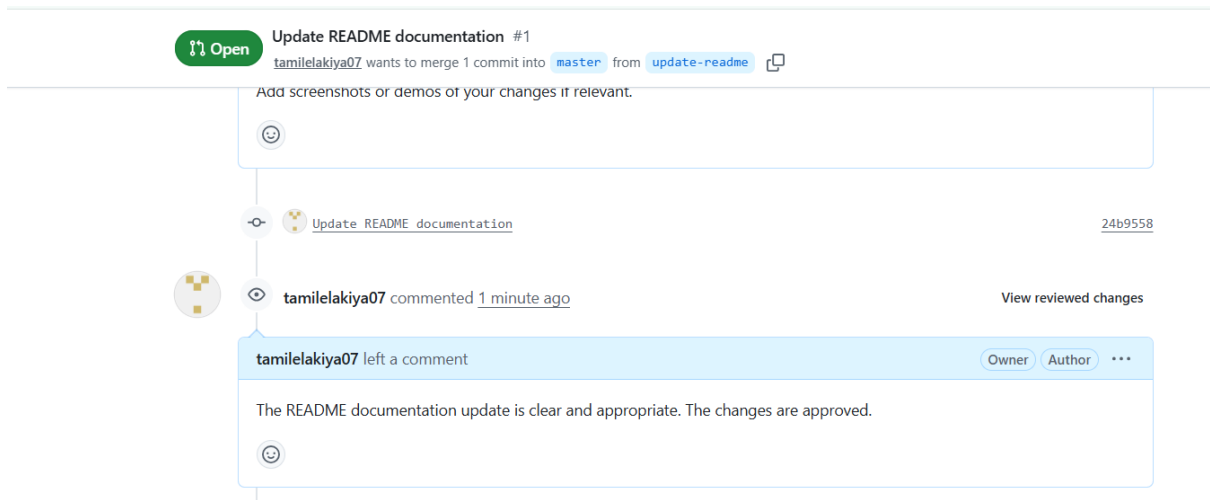
```
PS E:\Software engineering\Python> git status
On branch update-readme
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   README.md

PS E:\Software engineering\Python> git add .
PS E:\Software engineering\Python> git commit -m "Update README documentation"
[update-readme 24b9558] Update README documentation
 1 file changed, 11 insertions(+)
PS E:\Software engineering\Python> git push -u origin update-readme
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 529 bytes | 75.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
remote:
remote: Create a pull request for 'update-readme' on GitHub by visiting:
remote:   https://github.com/tamilelakiya07/Python/pull/new/update-readme
remote:
To https://github.com/tamilelakiya07/Python.git
 * [new branch]      update-readme -> update-readme
branch 'update-readme' set up to track 'origin/update-readme'.
```

7. Pull Request: Created a Pull Request from update-readme to the main branch.

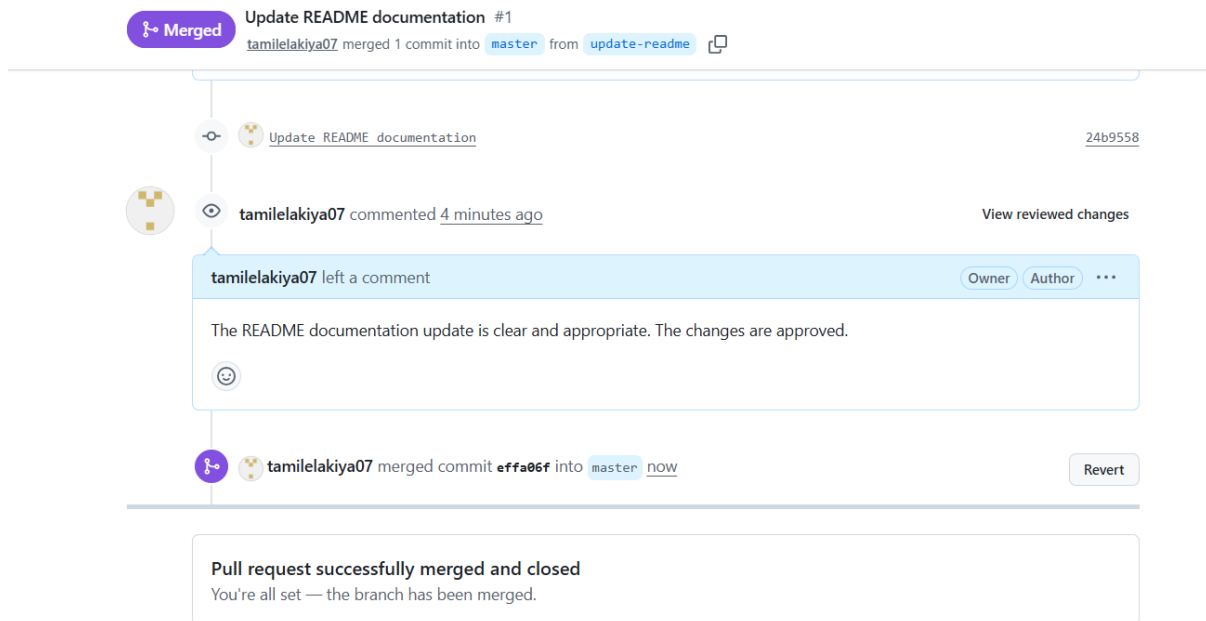


8. Review: Reviewed the changes and provided feedback through GitHub.



9. Response: Addressed feedback using additional commits and pushed the changes.

10. **Merge:** After approval, merged the Pull Request into the main branch.



Result

The fork-and-pull request workflow was successfully demonstrated. Changes were developed in a separate branch, reviewed through a Pull Request, updated based on feedback, and finally merged into the main branch.